

Inferred solution of Unsteady Stokes Flow using Physics Informed Gaussian Process

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Chapter 1

Introduction

1.1 Unsteady Stokes Flow

Behavior of fluid could be explain via the Navier Stokes Equation, which utilize Cauchy Momentum to do so. For an incompressible fluid,

$$\rho \frac{\partial u}{\partial t} + \nabla(u \cdot \nabla) = \nabla p + \eta \nabla^2 u + f_{ext} \quad (1)$$

For fluid with low Reynold number, the self-advection term could be neglected

$$\rho \frac{\partial u}{\partial t} = -\nabla p + \eta \nabla^2 u + f_{ext} \quad (2)$$

Unsteady Stokes flow is blah blah

1.2 Gaussian Process Regression

Gaussian Process or GP are defined by a series of infinite value that have a joint Gaussian distribution. A GP for function $f(x)$, with the input locations at $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]$, will provide values for $\mathbf{f}(\mathbf{X}) = [\mathbf{f}(\mathbf{x}_1), \mathbf{f}(\mathbf{x}_2), \dots, \mathbf{f}(\mathbf{x}_N)]$. The joint distribution is in the form of a multi-variate Gaussian

$$p(\mathbf{f}(\mathbf{X})|\mathbf{X}, \theta, \mu(\cdot), k(\cdot, \cdot)) = \mathcal{N}(\mu(\mathbf{X}), k(\mathbf{X}, \mathbf{X}; \theta)) \quad (3)$$

μ and k represent the average function and kernel function respectively

$$\mu(\mathbf{X}) = \langle f(\mathbf{X}) \rangle \equiv \langle f \rangle((x)) \quad (4)$$

$$k(X, X; \theta) = \langle \delta f(\mathbf{x}) \delta f(\mathbf{x}') \rangle \equiv \langle \delta f \delta f(\mathbf{x}, \mathbf{x}') \rangle \quad (5)$$

where θ is the hyperparameter of the kernel function. The kernel function used was a square exponential kernel.

$$k(x, x') = \exp\left(-\frac{(x - x')^2}{2l^2} + \gamma\right) \quad (6)$$

to capture the dimensionality of the problem, derived kernels were utilized such as product kernel, to calculate the kernel function.

$$k^\times(\mathbf{x}, \mathbf{x}') = k(x, x') \times k(y, y') \quad (7)$$

We can learn the function by inferring the value at each test positions $X_\star$. For a function f ,

$$\begin{bmatrix} \mathbf{f}^\star \\ \mathbf{f} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu^\star \\ \mu \end{bmatrix}, \begin{bmatrix} \mathbf{K}_{\star\star} & \mathbf{K}_\star \\ & \mathbf{K} \end{bmatrix}\right) \quad (8)$$

the distribution of $\mathbf{f}_\star \mid \mathbf{f} \sim \mathcal{N}(\mu + \mathbf{K}_\star \mathbf{K}^{-1} \delta \mathbf{f}, \mathbf{K}_{\star\star} - \mathbf{K}_\star \mathbf{K}^{-1} \mathbf{K}_\star^\mathbf{T})$

1.3 Integration of Physics

1.3.1 Main Idea

Using backward Euler we can express u^t as following

$$u^t = u^{t-1} + \frac{\Delta t}{\rho} (-\nabla p + \eta \nabla^2 u + f_{ext}) \quad (9)$$

Now,

$$u^t + \frac{\Delta t}{\rho} (\nabla p - \eta \nabla^2 u) = u^{t-1} + f_{ext} \quad (10)$$

The *LHS* of the equation, we defined as $\mathcal{G}$

$$\mathcal{G} \equiv u^t + \frac{\Delta t}{\rho} (\nabla p - \eta \nabla^2 u) \quad (11)$$

$$\mathcal{G} \equiv u^t + \frac{\Delta t}{\rho} f$$

$$\mathcal{G} \equiv u^{t-1} + f_{ext}$$

where f is

$$f = \nabla p - \eta \nabla^2 u \quad (12)$$

1.3.2 Prior

We can learn the function by inferring the value at each test positions $X_\star$. For a function f ,

$$\begin{pmatrix} \mathbf{u} \\ \mathbf{p} \\ \mathcal{G} \\ \mathbf{s} \end{pmatrix} \sim \mathcal{N}(0, \mathbf{K}) \quad (13)$$

$$K = \begin{bmatrix} \mathbf{K}_{uu} & \mathbf{K}_{up} & \mathbf{K}_{uG} & \mathbf{K}_{us} \\ & \mathbf{K}_{pp} & \mathbf{K}_{pG} & \mathbf{K}_{ps} \\ & & \mathbf{K}_{GG} & \mathbf{K}_{Gs} \\ & & & \mathbf{K}_{ss} \end{bmatrix} \quad (14)$$

1.3.3 Posterior

The prior of all the observed points and governing equation can be expressed as

$$\begin{bmatrix} u_b \\ S_L^1 \\ S_p^1 \\ \mathcal{G}^t \\ s \end{bmatrix} \sim \mathcal{N} \left(0, \begin{bmatrix} \mathbf{K}^{**} & \mathbf{Q} \\ \mathbf{Q}^T & \mathbf{K} \end{bmatrix} \right) \quad (15)$$

where, u_b denote the observed boundary value of the velocity. The correlation matrixes can be wrtitten as

$$\begin{bmatrix} \mathbf{K}_{uu} & \mathbf{K}_{up} & \mathbf{K}_{uG} & \mathbf{K}_{us} \\ & \mathbf{K}_{pp} & \mathbf{K}_{pG} & \mathbf{K}_{ps} \\ & & \mathbf{K}_{GG} & \mathbf{K}_{Gs} \\ & & & \mathbf{K}_{ss} \end{bmatrix} \quad (16)$$

The full posterior can be expressed as following, where $\mathcal{G}^{t-1}$ is defined as the *RHS* of the unsteady stokes equation.

$$\begin{bmatrix} u^t \\ p^t \end{bmatrix} \mid \begin{bmatrix} u_b \\ S_L^1 \\ S_p^1 \\ \mathcal{G}^t \\ s \end{bmatrix} \sim \mathcal{N}(\mathbf{q}^T)$$

Chapter 2

Methodology

Chapter 3

Results

1. $dt = 0.05$ and 5/10 loops work
 2. $dt = 0.01$ and 25 loops work (Testing the effect of big and small time interval) Should be fine
 2. $dt = 0.01$ and 25 and 200, 270, 340 datapoints work (Testing the effect training artificial data sizre) Should be fine

Note' right now the optimizer is optimizing the same function on the same value. At the start of each loop, the data should reset so a new hyperparam could be train on new data.